

5. Data Signal Flow

- (1). RF Signal Flow
 - * The photo detectors A1 - A6 output is applied to IC-101's RF IN INPUT (pin 39) via an RF amplifier within the optical pick-up.
 - * After being amplified by another RF amplifier within IC-101, the signal passes through an equalizer for 3T compensation, goes into the AGC (Auto Gain Control) circuit where it undergoes automatic gain control, and appears at the ARF (RF signal controlled by auto gain control) output (pin 35) of IC-101.
- (2). IF PLL Lock is lost, the frequency comparator's error output, this error signal will be controlled VCO's free-running frequency. The large error output of the frequency comparator drives the VCO until range. When the PLL is locked (TRON active), the FPC (frequency comparison error signal) is active and the VCO is controlled by the phase comparator's error output.

7. Control Logic (IC-102)

Commands used to set the servo circuit status are supplied from the system CPU in parallel 4 bit data (CNT1 - CNT4).

SENSE is the status read signal for IC-102, which is transferred to the system CPU. All other control logic operations are identical to those of the conventional control logic.

8. Focus Servo Circuit Operation

- (1). The outputs of the photo detector are coupled to pin 2 - 5 of IC-101, where they are phase corrected by the A1 + A2 and A3 + A4 signals and converted from current to voltage signals. They are transferred to the focus balance circuit and subjected to photo detector position error correction by means of VR101 (best eye adjustment).

- (2). The difference between the A1 + A2 and A3 + A4 signals, produced within the EF mixing circuit, serves as the Focus Error signal which appears at IC-101 pin 15. The signal is then applied to IC-102 pin 8. VR105 is used for offset voltage injection control.
- (3). Since the Focus switch is closed in the PLAY mode, the EF signal appears at IC-102 pin 9 and couples to the loop filter within IC-101, where it is used to determine the focus servo's frequency response. The output of the loop filter at IC-101 pin 23, is transferred to the focus driving circuit and drives the focus actuator.

6. PLL Operations

- (2). EFM Data Generation
 - * The RF signal applied to IC-102's ARF input (pin 12) is sliced by the DSL (data sliced) into a pulse signal EFM (eight to fourteen bit modulation), and then coupled to the PLL circuit.
 - * The PLL is synchronized with the EFM signal and then coupled to the signal processing block.

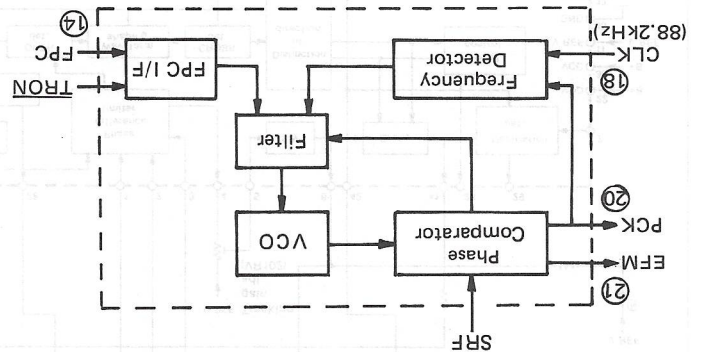


Figure 17-35 PLL Block

- (1). The new PLL circuit has replaced the PLL adjustment with a frequency comparator, one of whose input is clocked by the CLK signal (88.2KHz/11.34us) supplied from the digital signal processor.

The PLL circuit shows figure 17-35.

4. Servo Circuit Block Diagram

(Note) — P.B. signal

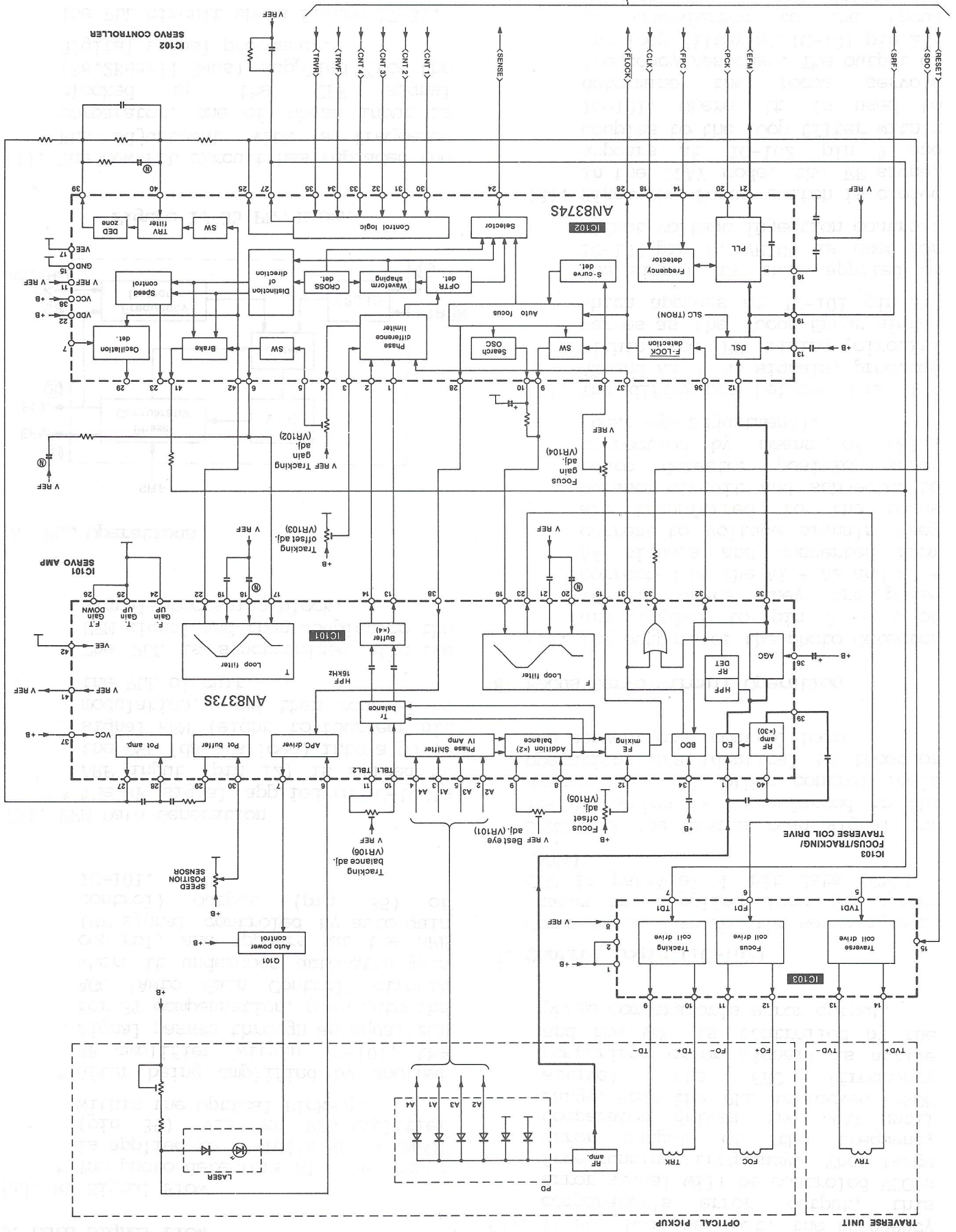


Figure 17-33 New Servo Circuit Block Diagram

9. Tracking Servo Circuit Operation

(1). The focus balance circuit output, A1 + A2 and A3 + A4, are also transferred to the tracking balance circuit. The output of the tracking balance circuit are applied to the phase difference limiter within IC-102, where they are phase compared to produce a tracking error signal.

The tracking error signal at IC-102 pin 4 is subjected to gain adjustment, then applied to the loop filter, within IC-101 via the tracking switch.

(2). The loop filter output appearing at IC-102 pin 22, is coupled to pin 42 (BR IN/ tracking error input) of IC-102, where it is fed into a braking circuit. The output of the braking circuit appears at pin 41 (BR OUT/ tracking drive control signal) and couples to the tracking coil driver to drive the tracking actuator's.

10. Traverse Servo Circuit Operation

(1). PLAY mode:

The tracking error signal applied to IC-102 pin 42 goes through the traverse switch and TRV (traverse circuit) filter (for response compensation) and appears at pin 39. This output is used to control the traverse actuator.

(2). KICK mode:

The KICK signal, decoded from the system CPU command by the control logic, appears at pin 27. It is used to control the tracking actuator which KICKs the pick-up by a span of 1, 10, or 100 tracks according to the system CPU command.

(3). Forced traverse mode:

The TRVF, TRVR signals supplied from system CPU to pins 34 and 35, respectively, are decoded to produce a traverse drive signal. The signal appears at pin 25 and drives the traverse coil.

(4). Speed sensor operation:

The varying voltage output of the potentiometer connected to IC-101 pin 30, is differentiated by the circuit at pin 27 to 29 and fed into the traverse coil. Smooth traverse action is achieved by feeding back a control signal which reflects the traverse movements of the optical pick-up. (The potentiometer yields a negative potential when the optical pick-up moves outward).

11. RF Signal Detection and Auto Focus Circuit

The auto focus circuit within IC-102 performs focus search. If it detects an RF signal S-curve it closes the focus switch contacts to close the focus servo loop. At this point the RF detector within IC-101 detects the RF signal and supplies it to the F.LOCK DET. input at IC-102 pin 36.

The F.LOCK signal is used to indicate to system CPU that focus is "IN". If focus goes "OUT" and the RF signal is no longer detected, the auto focus circuit opens the focus loop switch and starts focus search again.

12. Dropout Detection Circuit

(1). Black Band Dropout (BDO)

Dropouts are detected from the RF signal within IC-101. In order to stabilize the tracking servo against dropout, the BDO signal is supplied from IC-102 pin 37 to the phase difference limiter and used to hold the tracking servo operation.

(2). Signal Dropout (SDO)

If an RF signal dropout is detected by the RF detector, or a BDO is detected, the SDO signal is supplied to the signal processor to stabilize the CLV servo operation.

The SDO signal is supplied to the signal processor to stabilize the CLV servo operation. The SDO signal, produced within IC-102, is used to stabilize PLL circuit operation.

13. Braking Circuit

The braking circuit applies a brake to the tracking servo pull-in sequence required after a SKIP or KICK operation.

The direction of actuator movement is detected by the OFFR DET (off track detection), CROSS (tracking cross) and DIRECTION signals from IC-102 and a brake signal is added to the tracking error signal.

16. Pin Description of AN8373S

Pin No.	Mark	I/O Division	Function
1	AMP1	I	RF signal input (X30 amp.)
2	PDAD	I	Photo detector current input (A2)
3	PDA	I	Photo detector current input (A1)
4	PDBD	I	Photo detector current input (A4)
5	PDB	I	Photo detector current input (A3)
6	LPD	I	Non-inverting laser power input
7	LD	O	Laser power auto control output
8	FBL1	I	PD balance adjustment
9	FBL2	I	
10	TBL1	I	Tracking balance adjustment
11	TBL2	I	
12	FOFS	I	Focus offset adjustment
13	IVA	O	Current/voltage conversion output (A)
14	IVB	O	Current/voltage conversion output (B)
15	FE	O	Focus gain adjustment output
16	FPI	I	Focus error signal input
17	TP1	I	Tracking error signal input
18	C. TPL	I	Tracking error filter capacitor input
19	C. TPH	I	
20	C. FPL	I	Focus error filter capacitor input
21	C. FPH	I	

Pin No.	Mark	I/O Division	Function
22	TP0	O	Tracking error signal output
23	FPO	O	Focus error signal output
24	FGC	I	Focus gain up signal input (Not used, connected to GND)
25	TGC	I	Tracking gain up signal input (Not used, connected to GND)
26	GD	I	Focus/tracking down signal input (Not used, connected to GND)
27	PTO	O	Position detecting amp. output
28	PTI	I	Position detecting amp. input
29	PBO	O	Position detecting buffer output
30	POT	I	Position detecting buffer input
31	BDO	O	Dropout detection signal output
32	RFDET	O	RF detection signal output
33	SDO	O	Dropout detection pulse output
34	C. SBDO	I	Dropout detecting capacitor input
35	ARF	O	RF signal output
36	C. AGC	I	AGC detecting capacitor input
37	VCC	I	Power supply (+5 V input)
38	LDON	I	Laser power control input
39	RF IN	I	RF signal input
40	AMPO	O	RF signal output
41	VREF	O	Reference voltage output
42	GND	I	Ground terminal

18. Digital Circuit

The digital processing circuit is the heart of the CD player. It accepts EFM data transferred from the RF data punch circuit (IC-501: data slice & PLL) and provides EFM demodulation, data interleaving, error correction, and other signal processing functions. The following figure 18-1 shows a digital block diagram and correlations between the blocks.

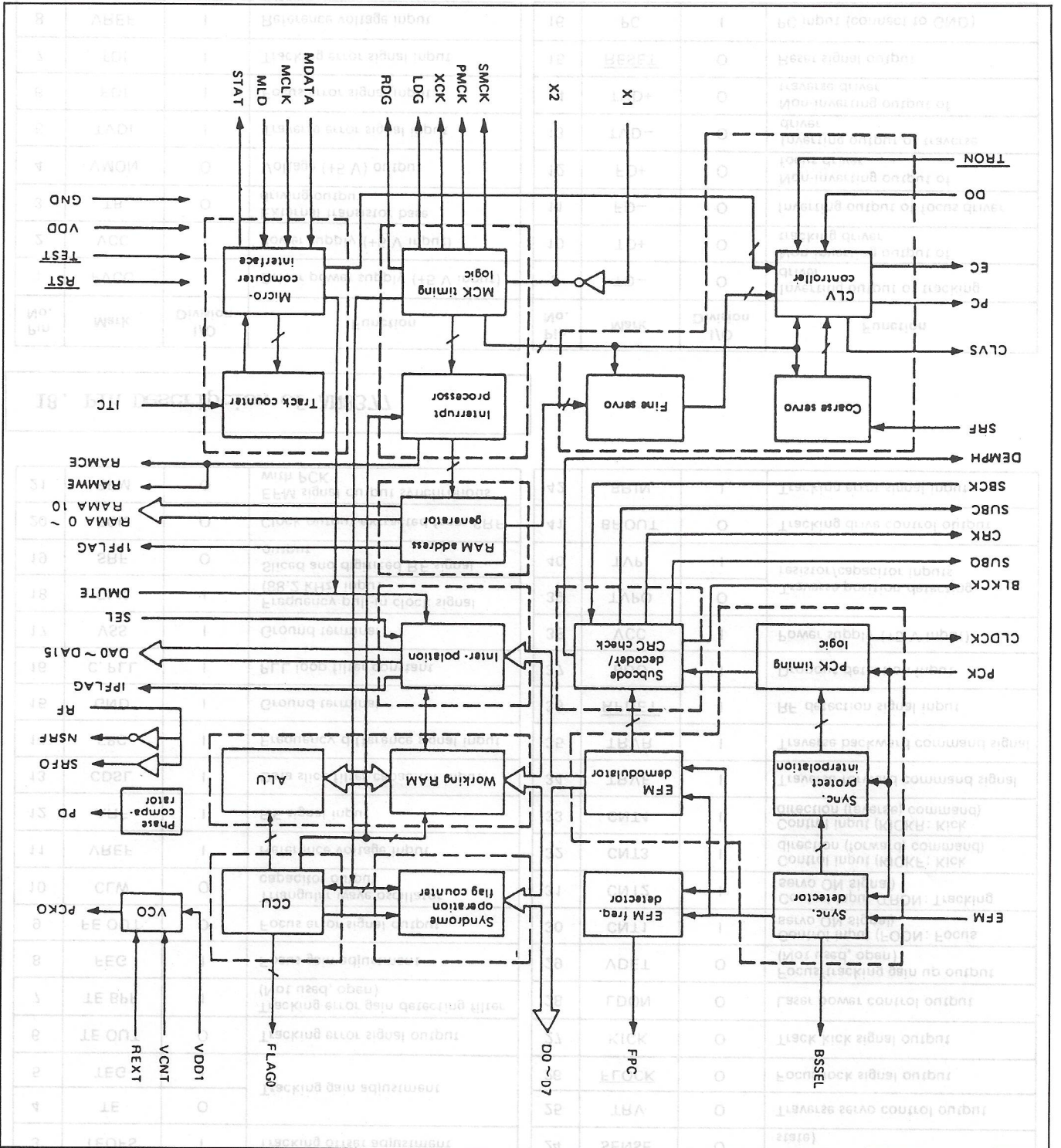


Figure 18-1 Block Diagram of MN6617

17. Pin Description of AN8374S

Pin No.	Mark	I/O Division	Function
1	LSA	I	Phase difference input (A)
2	LSB	I	Phase difference input (B)
3	TEOFS	I	Tracking offset adjustment
4	TE	O	Tracking gain adjustment
5	TEG	I	
6	TE OUT	O	Tracking error signal output
7	TE BPF	I	Tracking error gain detecting filter (Not used, open)
8	FEG	I	Focus gain adjustment
9	FE OUT	O	Focus error signal output
10	CLW	O	Triangular wave oscillator
11	VREF	I	Reference voltage input
12	ARF	I	RF signal input
13	CDSL	I	Data slice filter capacitor input
14	FPC	I	Frequency difference signal input
15	GND	I	Ground terminal
16	C. PLL	I	PLL loop filter constant
17	VSS	I	Ground terminal
18	CLK	I	Frequency pull-in clock signal (88.2 kHz) input
19	SRF	O	Sliced and digitized RF signal output
20	PCK	O	Clock output extracted from SRF
21	EFM	O	EFM signal output synchronous with PCK

18. Pin Description of AN8377

Pin No.	Mark	I/O Division	Function
22	VDD	I	Power supply (+5 V input)
23	SPCNT	O	Track crossing speed control output (Not used, open)
24	SENSE	O	Selector output (track crossing state)
25	TRV	O	Traverse servo control output
26	FLOCK	O	Focus lock signal output
27	KICK	O	Track kick signal output
28	LDON	O	Laser power control output
29	VDET	O	Focus/track gain up output (Not used, open)
30	CNT1	I	Control input (FOON: Focus servo ON signal)
31	CNT2	I	Control input (TRON: Tracking servo ON signal)
32	CNT3	I	Control input (KICKF: Kick direction (forward) command)
33	CNT4	I	Control input (KICKR: Kick direction (reverse) command)
34	TRVF	I	Traverse forward command signal
35	TRVR	I	Traverse backward command signal
36	RFDET	I	RF detection signal input
37	BDO	I	Dropout detection input
38	VCC	I	Power supply (+5 V input)
39	TVPO	O	Traverse position detecting resistor/capacitor inputs
40	TVPI	I	
41	BROUT	O	Tracking drive control output
42	BRIN	I	Tracking error signal input

Pin No.	Mark	I/O Division	Function
1	PVCC	I	Driver power supply (+5 V input)
2	VCC	I	Power supply (+5 V input)
3	TB	O	External transistor base driving output
4	VMON	O	Voltage (+5 V) output
5	TVDI	I	Traverse error signal input
6	FDI	I	Focus error signal input
7	TDI	I	Tracking error signal input
8	VREF	I	Reference voltage input

Pin No.	Mark	I/O Division	Function
9	TD-	O	Inverting output of tracking driver
10	TD+	O	Non-inverting output of tracking driver
11	FD-	O	Inverting output of focus driver
12	FD+	O	Non-inverting output of focus driver
13	TVD-	O	Inverting output of traverse driver
14	TVD+	O	Non-inverting output of traverse driver
15	RESET	O	Reset signal output
16	PC	I	PC input (connect to GND)

The jitter contained in the play signal is absorbed by the RAM. The basic internal operations of the digital processing circuit are explained on the following pages.

1) EFM Demodulation

(1) Sync signal detection, protection, and interpolation:

The EFM signal read off the disc (under optical system servo control) is coupled to the data slice/clock sampling circuit, where it is converted into a logic-level signal, separated into the EFM and PCK signals, and applied to the EFM demodulator.

The input EFM signal is first converted into NZRZI data (inverted for every one; non-inverted for every zero), then is applied to a 24 bits shift register, where the sync signal is extracted from the data.

Interpolation is required to cope with sync signal dropout or pseudo sync, signal generated due to defects on the disc. In order to prevent the synchronization system being disturbed by the pseudo sync, signal a small sync. Detection window is provided each time the sync signal should be found, so that sync detection is performed only within the window.

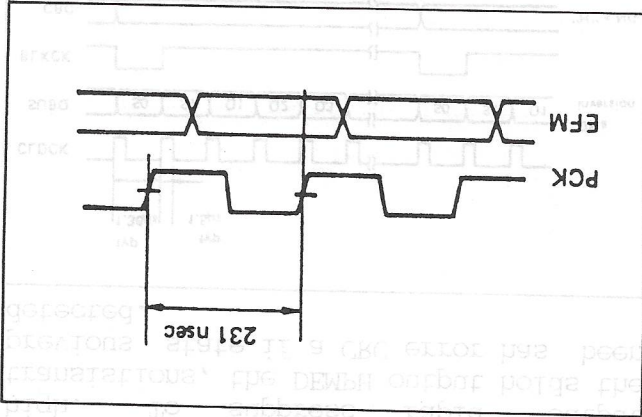
If no sync signal is detected in the window, interpolation is done by an internal 588 increment counter (which counts 588 PCK pulses). Also the number of sync detection failures is counted, and the width of the detection window is automatically controlled according to the count value.

The FCLV signal status changes depending on whether the sync signal has been detected or not:

*In every frame where on sync signal was detected, the FCLV is set low.

The PCK (bit period clock = 4.3218 MHz) is used to time signal processing during 16K RAM write. RAM read and subsequent operations are timed by a clock which is derived from the crystal-controlled master clock.

Figure 18-2 PCK and EFM Input Timing



*In every frame where the sync signal was detected, the FCLV is set high. (the sync detection failure counter is reset at a high transition of the FCLV)

**If a step out occurred, the RESY is set low.

**When the device attempts a synchronization, the RESY is set high.

The MN6617 can use the TRON (tracking on) signal for external synchronization control. When the tracking servo is off (TRON = high), the device always fully opens the window, so that it is ready for synchronization as soon as the tracking servo is turned on:

**When tracking servo is on, TRON is set low.

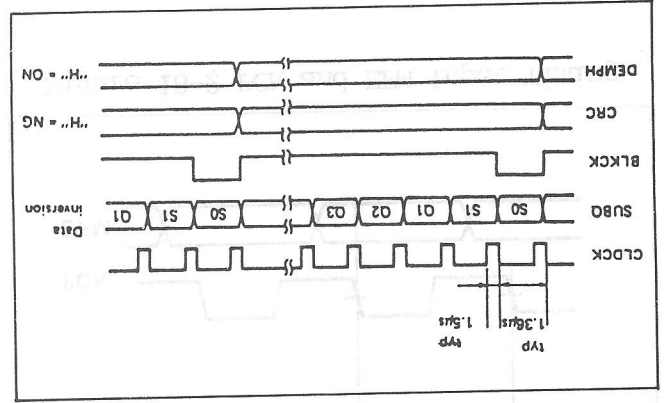
**When tracking servo is off, TRON is set high.

This enables reduced synchronization time when the player is accessing the next track.

(2) EFM demodulation

The output of the sync, detector's 24 bits shift register is coupled to a 14 to 8 decoder, where it is decoded into the original 8 bit data. Also subcodes S0 and S1 are decoded and transferred to the subcode decoder.

Figure 18-3 SUBQ, CRC and DEMPH Timing



The device extracts one bit, which indicates if emphasis is on or off from the Q data and supplies it to the DEMPH pin. De-emphasis is on when the DEMPH is high. To suppress rapid output transitions, the DEMPH output holds the previous state if a CRC error has been detected.

(4) De-emphasis detection

The 8 bits data, (P, Q, R, S, T, U, V, and W) are subjected to subcode demodulation and loaded into an 8 bits parallel to serial register at the CLK. It is then shifted out in series through the SUBC pin at an external SBCK shift clock timing.

(3) P to W data

Q data is subjected to CRC check, and the check result is output to the CRC and STAT pins of the device. If an error is detected, both CRC and STAT are set high, if no CRC error is detected, both CRC and STAT are set low. When the TRON input is high, the CRC check result output are forced high.

(2) CRC check on Q data

Q data is inverted and output as the SUBQ is synchronization with the CLK.

The BLCK signal indicating the block period is made active only if S0 and S1 are detected consecutively.

The device inputs the demodulated data from the subcode-block and sync signals (S0, S1) are transferred from the EFM demodulator with the frame clock (CLK).

(1) Q-data demodulation

2) Subcode Detector

The effective addresses for the 16k byte SRAM (static RAM) are given by an 11 bits read frame counter and an 11 bits write frame counter. The read frame counter is incremented by a crystal controlled frame clock. The write frame counter is incremented each time one frame (32 symbols) of EFM data is written into the RAM.

(1) RAM address generation

Each frame contains 2304 clocks of XCK cycle requires 12 XCK clocks (709 ns). So each frame contains up to 192 access slots.

To make these RAM accesses efficient, the RAM address controller assigns different levels of priority to RAM access. The level of priority is identified by the interrupt acknowledge logic so that the relevant address is supplied to RAM.

The MN617 accesses the external 16k byte RAM for the following read and write functions of the modulated EFM data:

- ** C1 unscramble
- ** Deinterleave
- ** C2 unscramble
- ** Error correction flags

3) RAM Interface

Figure 18-4 SUBC and SBCK Timing

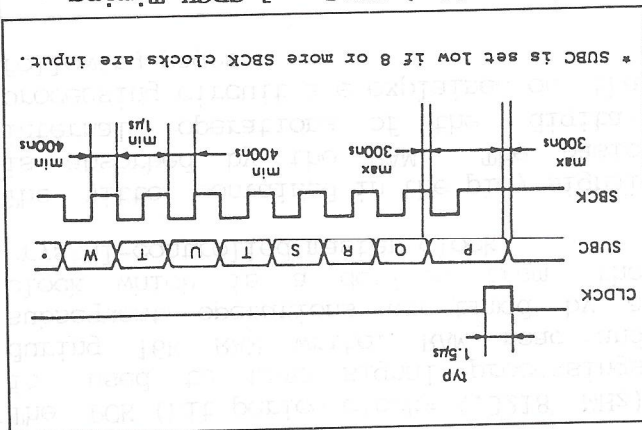


Figure 18-1

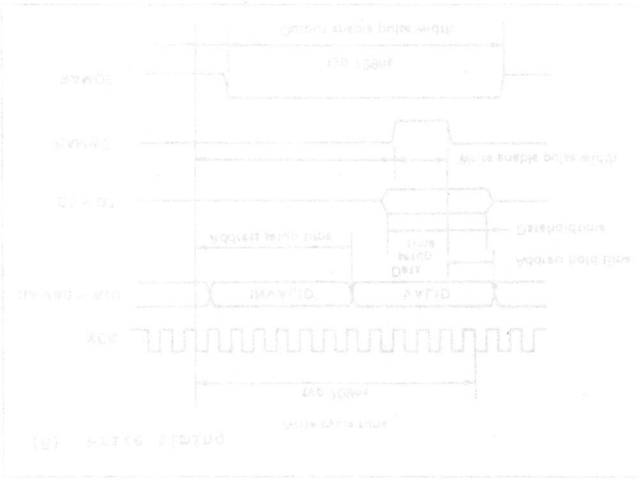
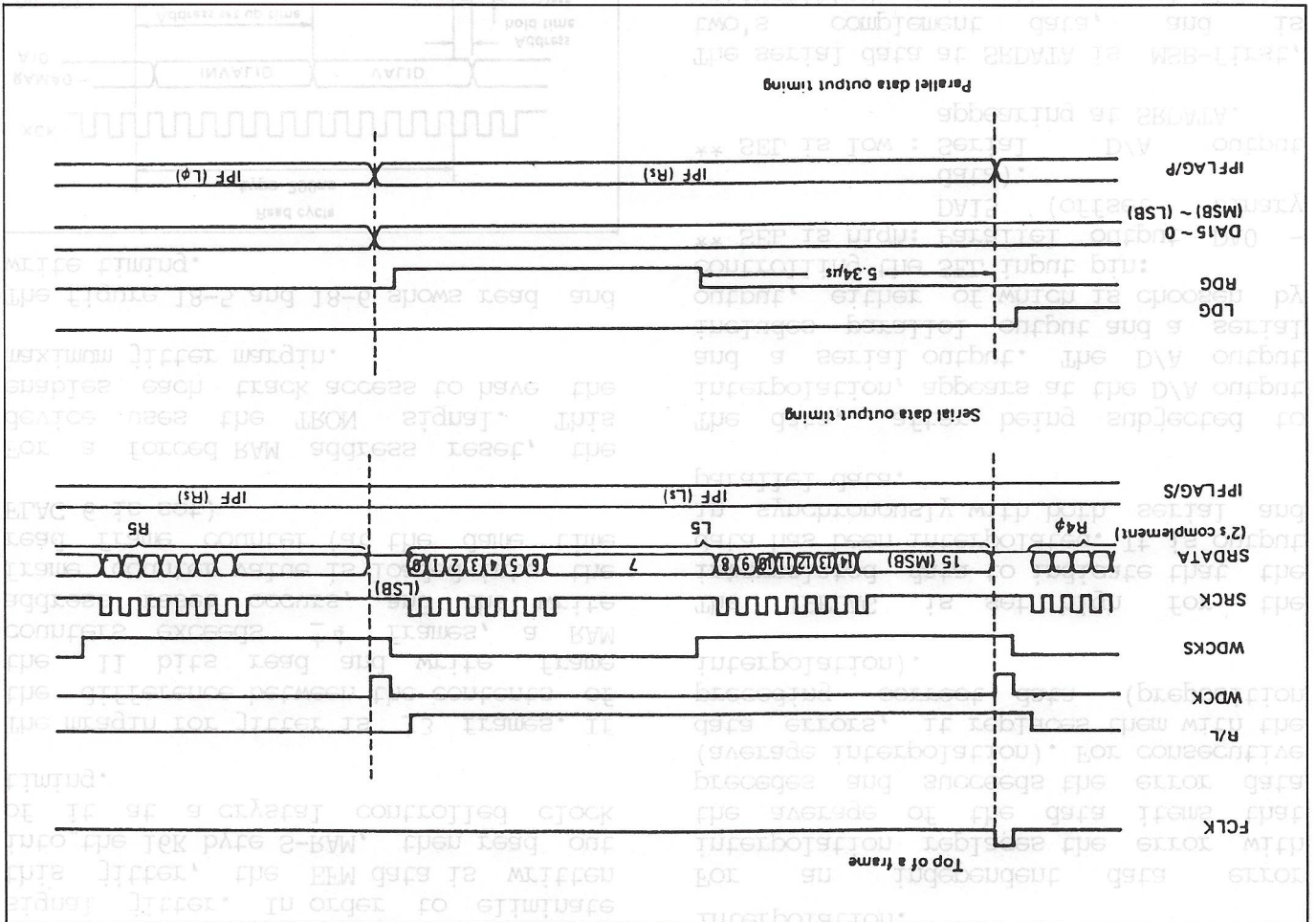


Figure 18-2



Figure 18-7



IPBYTE: Flag indicating the reliability of a symbol in SRDATA.
 BYTCK: Indicates the top of a symbol in SRDATA.
 WDCS: Flag indicating the reliability of a symbol in SRDATA.

Note
 The following pins are exclusively used for CD ROM.
 SRDATA: LSB-first, two's complement output.

(2) Jitter absorption and RAM address reset

The demodulated EFM data is synchronous with the PCK signal produced by the clock extraction logic and contains disc wow and flutter components that cause signal jitter. In order to eliminate this jitter, the EFM data is written into the 16K byte S-RAM, then read out of it at a crystal controlled clock timing.

The margin for jitter is ± 3 frames. If the difference between the contents of the 11 bits read and write frame counters exceeds ± 4 frames, a RAM address reset occurs, and the write frame counter value is loaded into the read frame counter (at the same time FLAG 6 is set).

For a forced RAM address reset, the device uses the TRON signal. This enables each track access to have the maximum jitter margin.

The figure 18-5 and 18-6 shows read and write timing.

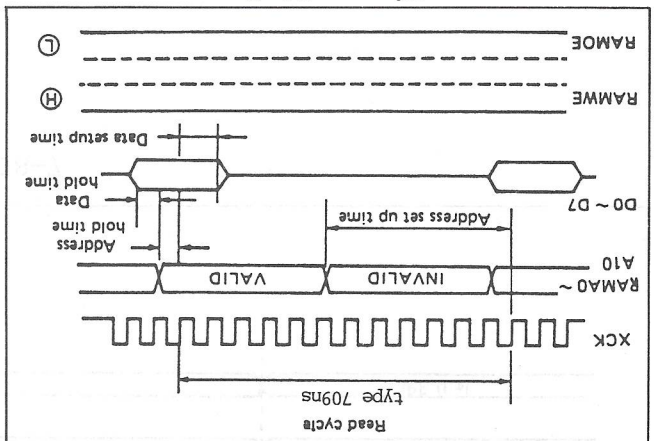


Figure 18-5

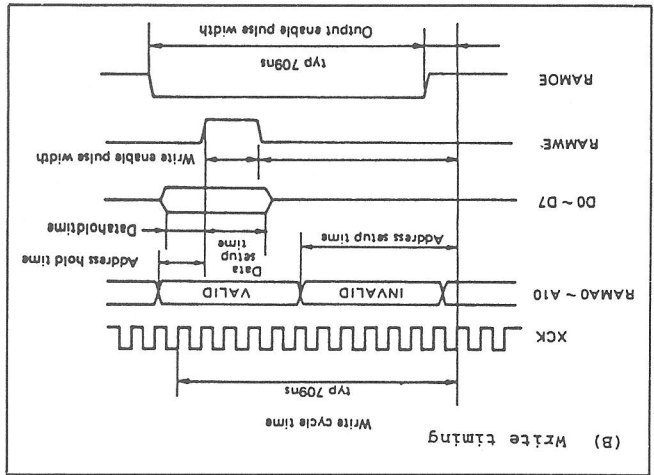


Figure 18-6

4) Interpolation and D/A Interface

Data is first subjected to error correction by the C2 decoder, the undergoes C2 unscramble. Incorrect data (16 bits) is subjected to error interpolation.

For an independent data error interpolation replaces the error with the average of the data items that precedes and succeeds the error data (average interpolation). For consecutive data errors, it replaces them with the preceding correct data (preposition interpolation).

The IPFLAG is set high for the interpolated data to indicate that the data has been interpolated. It is output in synchronously with both serial and parallel data.

The data, after being subjected to interpolation, appears at the D/A output and a serial output. The D/A output includes parallel output and a serial output, either of which is chosen by controlling the SEL input pin:

** SEL is high: Parallel output DA0 - DA15 (offset binary data).

** SEL is low: Serial D/A output appearing at SRDATA.

The serial data at SRDATA is MSB-first, two's complement data, and is interpolated under the control of the IPSEL input.

** IPSEL is low: Selects normal interpolation, in which both perposition and average are performed.

** IPSEL is high: All error data is interpolated by the L/R channel discrimination for the serial output is indicated by the R/L output pin.

** L/R is low: Left channel

** L/R is high: Right channel

Table 18-1

FLAG	5	4	3	2	1	Meaning
x	x	x	0	0	0	C1 - No error
x	x	x	0	0	1	C1 - Single error corrected.
x	x	x	1	0	0	C1 - Double error corrected.
x	x	x	1	1	0	C1 - Error in correctable.
0	0	0	0	0	0	C2 - No error
0	0	1	1	0	0	C2 - Single error corrected.
0	1	1	0	0	0	C2 - Double error corrected.
0	1	1	1	0	0	C2 - Double error in correctable.
1	0	0	0	0	0	C2 - Triple error corrected.
1	0	0	1	0	0	C2 - Triple error in correctable.

5) Error Correction

The error correction logic provides C1-double correction and C2-triple correction. C1 correction logic produces 4 types of C1 flags (no error, single correction, double correction, and triple or more correction). C2 correction logic selects either double or triple correction depending on the types and number of C1 flags and executes the selected correction. It also determines whether interpolation is

needed or not depending on the positions and types of C1 flags. If the logic has indicated that interpolation is needed, it transfers to the interpolation logic the C2 flag indicating which C1 flag is to be used for the interpolation. The interpolation logic selects the type of the C1 flag according to C2 flag, and provides interpolation only for the symbol for which the C1 flag is set. The error correction states can be monitored with flags 1 through 5.

6) Spindle Servo Control

The spindle servo control logic has four modes: CVL, 11T, 6T, and 6T break modes.

(1) 11T Mode

This mode is selected when the tracking servo is active (TRON = low) and the PLT is unlocked.

Since $XCK = 16.9344 \text{ MHz}$, the SRF frame sync signal pulse width is approximately

43XCK.

**Frame sync signal pattern:

= (1000000000100000000000000)

**Channel bit frequency/pulse width:

= $4.3218 \text{ MHz} / 0.32 \text{ ns}$

**XCK frequency/pulse width:

= $16.9344 \text{ MHz} / 0.059 \text{ ns}$

**XCK pulse count par 11T:

= $0.23 \times 11 / 0.059 = 43$

If no dropout exists in RF, 11T becomes the maximum pulse width. In the 11T mode, this maximum pulse width is determined by counting the XCK. The logic judges whether the obtained N11 is 43XCK or not and provides the following control functions:

** N11 is equal to 43XCK:

= EC is set to high impedance.

** N11 is greater than 43XCK:

= EC provides a low for every frame.

** N11 is smaller than 43XCK:

= EC provides a high for every frame.

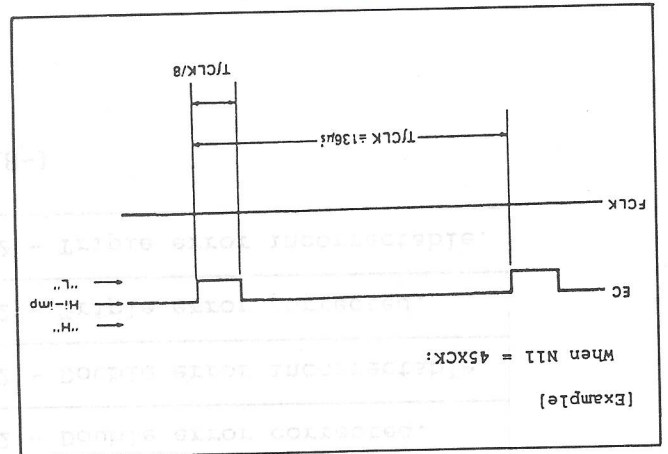


Figure 18-8

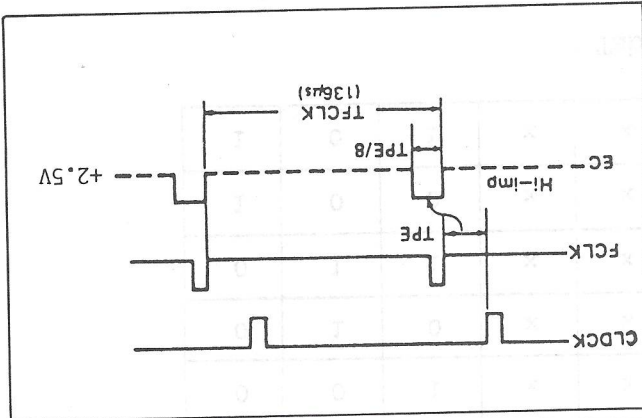


Figure 18-9

The EC signal's pulse width varies with the phase error, TPE (sec), between the CLCK and FCLK:

** If CLCK leads FCLK:

= EC is set low (acceleration)

** If CLCK lags behind FCLK:

= EC is set high (deceleration)

The device compares the FCLK's phase with CLCK's phase by comparing the value of the RAM read counter (which is synchronous with the reference frame sync signal (FCLK) with that of the RAM write counter (which is synchronous with the frame sync signal, CLCK, extracted from the EFM pattern), and provides a phase error signal at its EC pin.

(TRON = low).

provided the tracking servo is active CLV mode is automatically selected normal rotation rate. Either the 11T or rate is less than approx, 4.6% of the locked when the spindle motor rotation The CLV mode is selected if the PLT is

(2) CLV Mode

is controlled at an accuracy of approx, $\pm 1.16\%$ of the normal rotation rate.

When N11 = 43XCK, spindle motor rotation is controlled at an accuracy of approx, $\pm 1.16\%$ of the normal rotation rate.

TPECLK = $1/7350 = 136 \mu s$

EC's pulse width varies with the deviation of 43XCK (max.TPECLK/2).

Table 18-2

Mode		Address				D ₄	D ₃	D ₂	D ₁	Remarks	
		A ₄	A ₃	A ₂	A ₁						
CLV	0	0	0	1	FGARI L:W/FG H:TTON	H:TTON L:TTOFF	00 STOP Tree Run 01 ACC 10 Brake 11 No operation			-	
	Mute / ATT	0	0	1	0	-	ATT H: -12dB L: 0dB	MUTE H:mute L:0dB	-		
	T CAR K C O C R E T N U C	0	0	1	0	TC ₃	TC ₂	TC ₁	TC ₄	TC ₈	Low
		0	1	0	0	TC ₇	TC ₆	TC ₅	TC ₄	TC ₈	Middle
0		1	0	1	TC _B	TC _A	TC ₀	TC ₈	TC ₈	Middle	
0		1	1	0	Count enable	TC _E	TC _D	TC _C	TC _C	High	
S V T A T S	0	1	1	1	-	-	-	00 CRC 01 OTC 10 CLVS 11 TTSTOP	-		

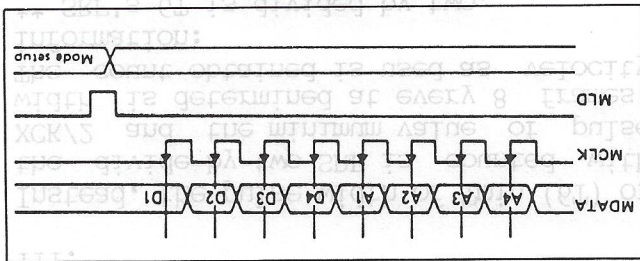
The various device modes can be set from the system CPU by inputting a 4 bits address and 4 bits data (both MSB first) signal into the device with the

(1) Signal Name

7) System CPU Interface

(6) Operation upon generation of dropout

- **The device ignores a dropout (DO) signal pulse width of less than 4.3µs.
- **If the DO pulse width exceeds 4.3µs:
 - 1.The EC is set to high impedance (free run) when in high impedance
 - 2.The EC is set to high impedance (free run) when in CLV mode.
 - 3.DO is ignored when in 6T mode.
 - 4.Brake is not released when in 6T brake mode.



If off-tracking occurs, the device does not operate normally in any mode other than the 6T mode. It must be placed in the free-run mode by the DO signal or system control command.

(7) Operation upon off tracking

Following timing of the MDA7A, MCK, and MLD lines as shows figure 18-12.

Figure 18-12

(3) 6T Mode

The 6T mode is used for coarse servo control for high-speed access. During traverse head movement, the 11T mode is not available since the RF signal is shut down for an interval of more than 11T.

Instead, the pulse width of T_{min} (6T) of the divide-by-two SRF is counted with $XCK/2$ and the minimum value of pulse width is determined at every 8 frames. The count obtained is used as velocity information:

** SRF's 6T is divided by two.
 ** $XCK/2 = 0.059 \text{ ns} \times 2 = 0.118 \text{ ns}$
 ** 6T's pulse width = $0.23 \text{ ns} \times 6 = 1.38 \text{ ns}$
 ** If 6T is counted by $XCK/2$:
 $= 1.38 \text{ ns} / 0.118 \text{ ns} = 11.7 \text{ counts}$

Since 6T's pulse width is nearly 11 x $XCK/2$, the EC output provides:
 ** A low (acceleration) if N6 (obtained count) is greater than 11.
 ** A high (deceleration) if N6 is smaller than 11.
 In any other case, the EC is set to high impedance.

6T divided by two:

Example
 When 6T's pulse width = $12 \times XCK/2$

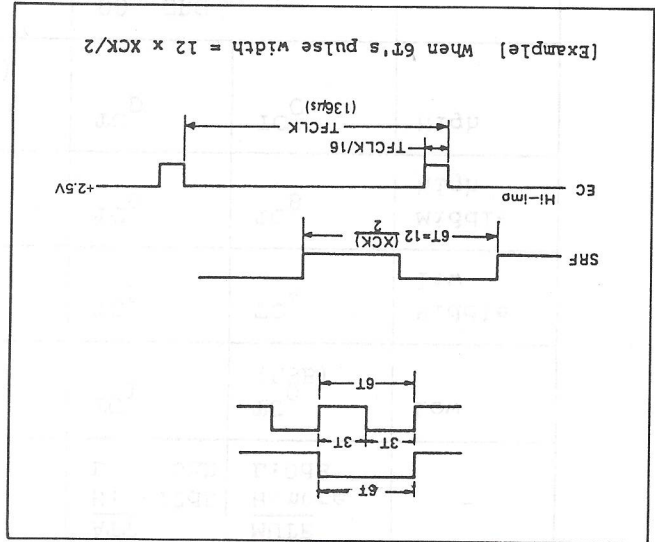


Figure 18-10

(4) 6T Brake Mode

The 6T brake mode is used to stop the turntable rotation. The device enters this mode when it receives a TTOFF command from system IC-401 by MDATA pin 59.

The device first places its EC pin in the forced deceleration mode. When the minimum pulse width (6T) of the divide-by-two SRF is reduced to approx. 7 times that at the normal rotation rate, the device sets the EC output to high impedance and set the PC output high.

If a dropout occurs, the device does not select brake release, so that the turntable continues decelerating. A dropout with a duration of less than 4.3 µs is however ignored. The figure 18-11 shows 6T brake mode.

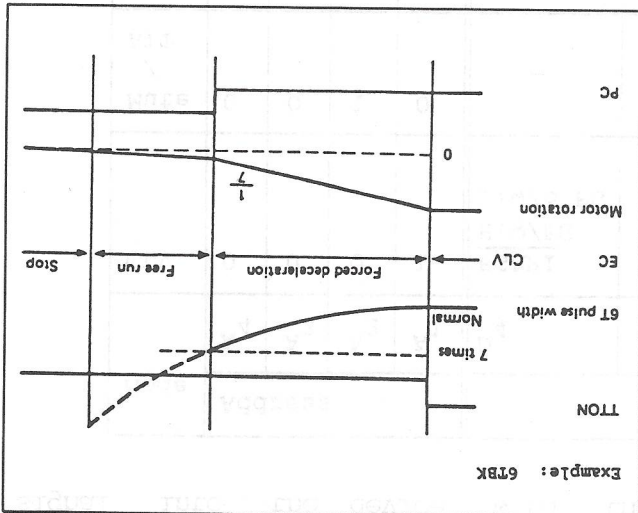


Figure 18-11

(5) Speed Limiter

When the player uses a motor with FG, the user can let the device control the upper and lower limiter of motor speed by applying the FG signal to the FG input of the device.

When it is more than 77 Hz, the EC pin is set high (forced deceleration).

If the FG generates 6 pulses per rotation:

** EC provides forced acceleration at 140 rpm or less.
 ** EC provides forced deceleration at 770 rpm or more.

Table 18-4

D2 D1	Signal name	Contents
0 0	CRC	Result of the CRC H: NG L: OK
0 1	TTC	Track Counter Output
1 0	CLVS	Spindle Servo Phase Sync. H: CLV L: Coarse servo
1 1	TTSTOP	Stop signal TURNTABLE H: Rotating; L: Stationary

The DMUTE pin may be used for direct muting since it is logically OR'ed with the MUTE signal.

This command allows the D/A output data to be muted in case of dropout. The D/A output is muted to logical all zeros when MUTE is set high, muting is cleared when it is set low (only when DMUTE=low)

(e)MUTE

This command allows the D/A output data to be attenuated. It is effective for both parallel DA0 - DA15 output and serial SRDATA output.

Output is attenuated -12dB when ATT is set high. And attenuation is released when ATT is set low.

(d)ATT

Table 18-3

D2 D1	Mode	Operation
0 0	STOP	Turntable forced free-run
0 1	ACC	Turntable forced acceleration
1 0	Brake	Turntable forced deceleration
1 1		No Operation

(c)Forced Turntable Control Mode

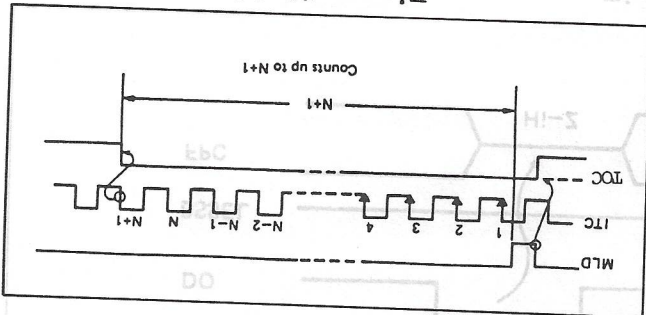
This command is used for starting and stopping turntable operation:
** TTON is set high = Turntable starts.
** TTON is set low = turntable stops.

(b)TTON

The MN6617 has a motor speed limiter which uses the FG signal to be compatible also with motors having no FG, the device can deactivate the speed limiter by setting FG ARI is low.
**FG ARI is set high:
= FG speed limiter is activated.
**FG ARI is set low:
= FG speed limiter is deactivated.

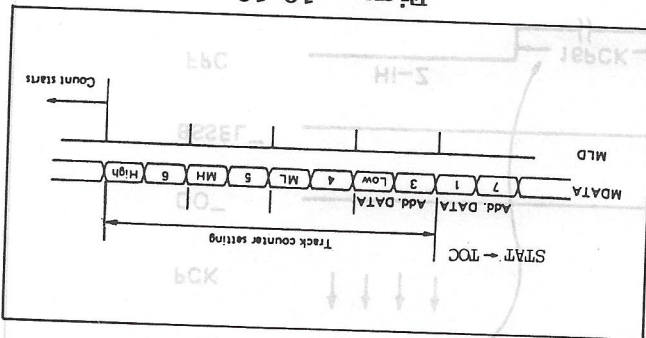
(a)FG ARI

Figure 18-14



TCO output timing

Figure 18-13

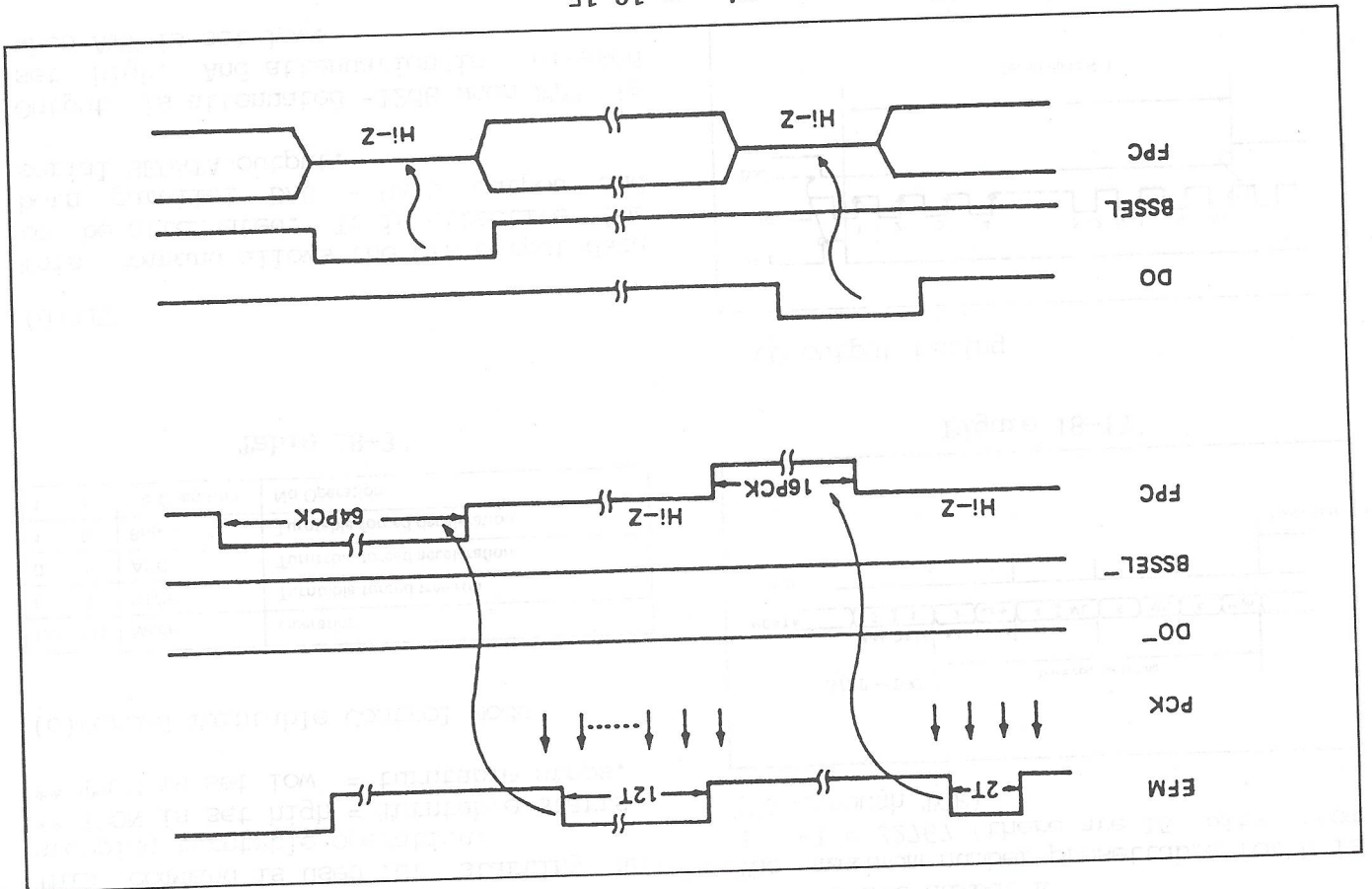


How to set number N
The maximum number presettable for N is 2 -1 = 32767 (there are 15 bits from TCO through TCE)

(f)Track Counter

When an appropriate number N (the number of tracks to be counted, which is computed by the CPU) is furnished to the TCO-TCE inputs and the count enable pin is set high, the device counts the negative edge of the clock applied to the TTC pin (Track Counter Input). The TCO (internal) is set low when the clock count reaches N+1.

Figure 18-15



8) FPC Interface

The FPC interface compares the EFM signal frequency with the PCK frequency (extracted PLL clock)

(a). If PCK frequency is higher than EFM frequency the FPC is set to high.

(b). If PCK frequency is lower than EFM frequency the FPC is set to low.

(c). If the two frequencies are identical the FPC is set to high impedance.

The EFM signal pulse width is counted using the PCK, and when 2T is detected, the FPC provides a low for the 16 PCK pulses interval.

When 12T is detected, the FPC provides a high for 64 PCK pulses interval.

If a dropout sense signal (DO = high) is received, the FPC is held at high impedance even while synchronization is maintained (in which BSSEL is high).